Jiatong Gao

510-309-2108 | jgao82@jhu.edu | U.S. Permanent Resident | sites.google.com/view/jiatonggao

Education

Johns Hopkins University Aug 2025 – Dec 2026 (expected)
M.S.E. in Applied Mathematics and Statistics (GPA: 3.76)

- **Relevant Coursework:** Probability Theory, Bayesian Statistics, Optimization for Data Science, Probabilistic Machine Learning, Generative AI & Agentic Systems for Clinical Data Science & Biopharmaceutical Statistics

University of California, Davis Sep 2022 – June 2025
B.S. in Applied Mathematics, Minor in Statistics, Graduated with Honors (GPA: 3.90)

Experience

Graduate Research Assistant, Computational Genomics & Single-Cell Trajectory Inference Oct 2025 – Present
Ji Lab (Prof. Hongkai Ji, Dept. of Biostatistics); Johns Hopkins Bloomberg School of Public Health *Baltimore, MD*

- **Developing Master's thesis** on **probabilistic single-cell trajectory inference** using patient-specific **TCR clonotypes** as clonal anchors, advised by Prof. Holden Lee (Dept. of Applied Mathematics and Statistics)
- **Implementing graph reconstruction** with rooted shortest paths, MSTs, and **KMB Steiner trees** on a 155,000+ T-cell **kNN graph** in Harmony space
- **Building an MCMC framework** over feasible Steiner trajectories to quantify uncertainty across alternative immune-state connections
- **Validating algorithms** with hand-derived graphs and external **Stanford / SteinLib benchmarks**
- **Analyzing Long COVID PBMC data** from 16 patients with longitudinal **scRNA-seq + TCR clonotype** integration
- **Reintegrating, reclustering, and annotating T/B cells** using `Scanpy`, `Harmony`, `Seurat`, `CellTypist`, RNA markers, and ADT profiles
- **Collaborating with JHU Medicine** on T-cell state progression, clonotype-supported trajectories, and biologically plausible bridge states
- **Building reproducible Python/R pipelines** with `AnnData`, `pandas`, `SciPy`, `Scanpy`, `Seurat`, and **SLURM/HPC**

Computational Genomics Researcher, Genetic Perturbation Modeling May 2025 – Aug 2025
NASA Open Science Analysis Working Groups (AWG), AI/ML AWG *San Francisco Bay Area*

- **Processing Perturb-seq transcriptomic data** across **2,679 observations and 8,749 genes** with `Scanpy`, `AnnData`, `NumPy`, and `pandas`
- **Evaluating gene-expression perturbation structure** with QC, PCA, feature-loading analysis, and model-validation workflows for **single-cell and bulk RNA-seq**

Data Science Research Intern Sep 2023 – Mar 2024
Department of Mathematics, University of California, Davis *Davis, CA*

- **Deriving and applying SVD** for dimensionality reduction and feature interpretation in the Enron email corpus
- **Analyzing communication among ∼150 employees** with NLP preprocessing and presenting results at the UC Davis Winter 2024 DRP Conference

Projects

TriAlign: Clinical Trial Eligibility Representation Learning | *Generative AI for Clinical DS* Fall 2026

- **Developing clinical-criterion retrieval and comparison workflows** across trial protocols to distinguish clinical meaning from surface-level textual similarity
- **Combining dense embeddings, structured comparison rules, and LLM judgments** to classify equivalence, restrictiveness, overlap, contradiction, and ambiguity
- **Constructing hard-negative training pairs** from reviewed eligibility criteria to improve encoder discrimination while preserving directional clinical relationships

- **Evaluating reliability** with leakage-resistant trial splits, human-reviewed labels, calibration, uncertainty, and abstention to identify cases requiring **human review**; using `PyTorch`, `FAISS`, `Pydantic`, and Python

Type 2 Diabetes Risk Modeling with Self-Collectable Survey Data | *Health DS* Fall 2026

- **Modeling 12-month T2D risk** from self-collectable demographic, anthropometric, and survey features in the **All of Us Research Program** under NIH
- **Fitting ridge-penalized logistic regression** with restricted cubic splines for nonlinear age, BMI, and waist-circumference effects
- **Optimizing screening thresholds** to minimize screening burden subject to prespecified sensitivity constraints
- **Evaluating clinical utility and reliability** against ADA and USPSTF criteria using ROC-AUC, PR-AUC, calibration, Brier score, sensitivity, specificity, and PPV; selecting regularization by **5-fold cross-validation**

Single-Cell Representation Learning: scGPT vs. Sparse Dictionary Learning | *ML & Agent* Fall 2026

- **Comparing single-cell representations** from pretrained **scGPT embeddings**, sparse dictionary learning, and processed gene-expression features on public **scRNA-seq** data
- **Training regularized multinomial logistic regression** on matched representations and evaluating **Macro-F1, balanced accuracy, and log loss**
- **Analyzing interpretable gene programs** through sparse dictionary atoms, high-weight genes, and cell-type activation patterns
- **Building reproducible Python pipelines** with `PyTorch`, `AnnData`, `NumPy`, `pandas`, scGPT, and Git-based collaborative workflows

Technical Skills

Programming — Python, R, SQL, MATLAB, LaTeX
GenAI & LLM Systems — LLM applications, agentic workflows, prompt design, dense embeddings, retrieval, hard-negative construction, model evaluation, calibration, abstention, `FAISS`, `Pydantic`
Single-Cell & Computational Genomics — scRNA-seq, PBMC analysis, TCR/BCR clonotypes, immune-lineage reintegration and reclustering, single-cell trajectory inference, graph-based lineage modeling, marker-based cell annotation, RNA/ADT integration
Single-Cell Tools — `Scanpy`, `AnnData`, `Seurat`, `Harmony`, `CellTypist`, Leiden clustering
Python & Scientific Computing — `NumPy`, `pandas`, `SciPy`, `h5py`, `matplotlib`, sparse matrices, graph algorithms
Statistical & ML Methods — Bayesian modeling, MCMC, probabilistic modeling, logistic regression, gradient boosting, stochastic optimization, dimensionality reduction
Bioinformatics & Statistical Analysis — single-cell quality control, cluster marker analysis, differential-expression workflows, lineage-specific subclustering, clonotype integration, trajectory reconstruction
Computing — SLURM, Linux, JHU high-performance computing, Git/GitHub, reproducible Python/R pipelines
R — `Seurat`, `limma`, `edgeR`
Certifications — Responsible Conduct of Research; Social & Behavioral Research in Human Subjects (CITI Program)

Professional & Research Affiliations

JHU Medicine Long COVID & Immune Genomics Collaboration Fall 2025 – Present
Ji Lab collaboration with JHU Medicine investigators, including the Andrea Cox Lab

- Participate in cross-disciplinary review of longitudinal PBMC and immune-lineage analyses, connecting computational single-cell results with T-cell biology, clonotype behavior, longitudinal sampling, and biologically plausible cell-state transitions

JHU Medicine Bloomberg–Kimmel Institute for Cancer Immunotherapy Fall 2025 – Present
Joint Single-Cell Genomics Research Meetings through Ji Lab collaborations

- Participate in joint computational and experimental single-cell research discussions spanning immune-state annotation, transcriptomic interpretation, and translational immunology

NASA Open Science Data Repository (OSDR) Analysis Working Groups Summer 2025 – Present
AI/ML Analysis Working Group, Genetic Perturbation Modeling Group